Deep learning project

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2026-05-02

Introduction

Motor insurance pricing relies heavily on accurately modeling claim frequency in order to assess risk and set appropriate premiums. In this context, fleet insurance policies present specific challenges due to the heterogeneity of insured entities and the presence of sparse claim data.

The objective of this project is to model the number of claims (`nombre_de_sinistre`), treated as a count variable, while properly accounting for exposure (`Exposition_au_risque`). This problem falls within the framework of frequency modeling for insurance risk.

To address this task, a comprehensive modeling framework is implemented. A Generalized Linear Model (GLM) with a Poisson distribution is first introduced as a benchmark, due to its interpretability and strong theoretical foundations in actuarial science. More flexible approaches, such as neural networks, are then explored to capture potential nonlinear relationships and interactions between variables.

In addition, unsupervised learning techniques—including autoencoders and clustering methods—are investigated to uncover latent structures in the data and provide alternative segmentation of the portfolio. Model interpretability is further analyzed using tools such as Partial Dependence Plots (PDP), Individual Conditional Expectation (ICE), LIME, and SHAP, allowing for both global and local understanding of model behavior.

Overall, this study aims to evaluate the trade-off between interpretability, predictive performance, and segmentation capability across different modeling approaches in the context of motor insurance risk.

Data preprocessing

Exploratory analysis

The dataset is first loaded and explored to understand its structure and assess overall data quality. This initial step involves examining variable types, checking for missing values, and reviewing summary statistics for numerical features.

Table 1: Summary of dataset descriptive statistics

	Variable	Type	Missing	mean	std	min	max
0	Unnamed: 0	int64	0	55520.55	32391.64	2.00	120000.0
1	ANNEE	int64	0	2012.05	0.81	2011.00	2013.0
2	Date_Deb_Situ	object	0	NaN	NaN	NaN	NaN
3	Date_Fin_Situ	object	0	NaN	NaN	NaN	NaN
4	Classe_Age_Situ_Cont	object	0	NaN	NaN	NaN	NaN
5	Creation_Entr	int64	0	1.12	0.33	1.00	2.0
6	Mode_gestion	object	0	NaN	NaN	NaN	NaN
7	Zone	int64	0	2.64	1.27	1.00	6.0
8	Segment	int64	0	1.49	0.50	1.00	2.0
9	Fractionnement	object	0	NaN	NaN	NaN	NaN
10	Exposition_au_risque	float64	0	0.44	0.32	0.08	1.0
11	Age_du_vehicule	object	0	NaN	NaN	NaN	NaN
12	franchise	object	0	NaN	NaN	NaN	NaN
13	Nombre_vehicule	int64	0	1.00	0.00	1.00	2.0
14	Valeur_assuree	object	0	NaN	NaN	NaN	NaN
15	FORMULE	int64	0	3.36	0.72	2.00	5.0
16	Categorie_ensemble	int64	0	1.00	0.00	1.00	1.0
17	Num_contrat	int64	0	8241.83	5193.01	1.00	20994.0
18	Activite	int64	0	4.34	2.52	1.00	8.0
19	Type_Apporteur	int64	0	1.37	0.62	1.00	3.0
20	ValeurPuissance	int64	0	6.23	1.97	1.00	11.0
21	nombre_de_sinistre	int64	0	0.05	0.24	0.00	7.0
22	IMMAT	int64	0	40218.76	23917.87	2.00	89744.0

This analysis reveals that there are no missing values in the dataset, which eliminates the need for imputation and helps preserve model accuracy. At the same time, several variables such as `Age_du_vehicule`, `Fractionnement`, `Franchise` and `Valeur_assuree` are stored as categorical (object) types. This indicates that a substantial portion of the dataset consists of categorical information that will require appropriate encoding before modeling.

The target variable, `nombre_de_sinistre`, has a mean of 0.05 and a maximum value of 7, indicating that it is a count variable with a highly skewed distribution dominated by zero claims. This confirms that the dataset is sparse and that the modeling task falls within the context of count data analysis. Another key variable is `Exposition_au_risque`, which represents the time at risk. It has an average value of approximately 0.44 and ranges from 0.08 to 1. This variable plays a critical role in frequency modeling, as it allows the number of claims to be adjusted according to the duration of exposure.

Feature Engineering

To improve the predictive performance of the model, several feature engineering steps are applied.

First, date variables are converted into a standard datetime format. From these, more informative features are extracted, including the contract start year (`year_start`), contract start month

(`month_start`) and contract duration in days (`duration_days`). These engineered variables allow the model to capture temporal patterns such as seasonality and contract length effects. The original date variables (`Date_Deb_Situ` and `Date_Fin_Situ`) are subsequently removed, as their information is fully represented by these new features.

Next, numerical transformations are performed. The insured value (`valeur assuree`) is converted into a numeric format and log-transformed into `log_valeur_assuree` o reduce skewness and stabilize variance. Similarly, vehicle age (`Age_du_vehicule`) is converted into a usable numeric variable (`vehicule_age_num`)-

Feature selection is then carried out to retain only relevant predictors. Identifier variables such as `Unnamed 0`, `INMAT` and `Num_contrat` are removed, as they do not contain predictive information. Variables with little or no variability (e.g `Nombre_vehicule` and `Categorie_ensemble`) are also excluded, since they do not contribute to model performance.

In addition, redundant variables are removed once they have been replaced by transformed versions. For instance, `valeur assuree` is replaced by `log_valeur_assuree`, and `Age_du_vehicule` by `vehicule_age_num`. Similarly, the original year variable (`ANNEE`) becomes redundant after extracting detailed temporal features from the contract start date.

The exposure variable (`Exposition_au_risque`) is treated separately, as it is incorporated as an offset in the modeling stage rather than used as a standard predictor.

Encoding and Data Splitting

Before modeling, categorical variables are converted into numerical form using one-hot encoding. The dataset is then divided into a training set and a validation set, using an 80/20 split. Care is taken to ensure consistency between the two datasets by aligning the validation set with the feature space defined by the training data.

At the end of this process, the dataset is fully prepared for modeling, with cleaned variables, engineered features, and clearly defined target and exposure variables.

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1. Supervised Learning

1.1 Benchmark model

Because we want a benchmark model, a poisson GLM is fitted. An intercept is included, and the exposure variable is incorporated as an offset to properly model claim frequency. Since the logarithm of exposure is used, zero values are adjusted to avoid numerical issues.

Dep. Variable:	nombre_de_sinistre	**No. Observations:**	69782
Model:	GLM	**Df Residuals:**	69758
Model Family:	Poisson	**Df Model:**	23
Link Function:	Log	**Scale:**	1.0000
Method:	IRLS	**Log-Likelihood:**	-13919.
Date:	Sat, 02 May 2026	**Deviance:**	20805.
Time:	17:23:00	**Pearson chi2:**	7.49e+04
No. Iterations:	7	**Pseudo R-squ. (CS):**	0.004069
Covariance Type:	nonrobust		

	coef	**std err**	**z**	**P> |z|**	**[0.025**	**0.975]**
const	43.3811	41.834	1.037	0.300	-38.611	125.373
Creation_Entr	-0.1873	0.057	-3.309	0.001	-0.298	-0.076
Zone	0.1037	0.013	7.944	0.000	0.078	0.129
Segment	-0.0193	0.037	-0.526	0.599	-0.091	0.053
FORMULE	-0.1047	0.026	-3.969	0.000	-0.156	-0.053
Activite	-0.0112	0.007	-1.605	0.108	-0.025	0.002
Type_Apporteur	-0.0951	0.028	-3.356	0.001	-0.151	-0.040
ValeurPuissance	0.0937	0.009	10.605	0.000	0.076	0.111
year_start	-0.0225	0.021	-1.083	0.279	-0.063	0.018
month_start	-0.0109	0.007	-1.557	0.119	-0.025	0.003
duration_days	-0.0012	0.000	-5.964	0.000	-0.002	-0.001
log_valeur_assuree	4.379e-15	5.54e-15	0.790	0.429	-6.48e-15	1.52e-14
vehicule_age_num	-3.356e-15	4.32e-15	-0.776	0.438	-1.18e-14	5.12e-15
Classe_Age_Situ_Cont_2 - 3 ans	-0.0571	0.059	-0.969	0.332	-0.172	0.058
Classe_Age_Situ_Cont_3 - 4 ans	-0.0227	0.060	-0.377	0.706	-0.141	0.095
Classe_Age_Situ_Cont_4 - 5 ans	-0.0216	0.061	-0.351	0.725	-0.142	0.099
Classe_Age_Situ_Cont_<= 1 ans	-0.0900	0.054	-1.654	0.098	-0.197	0.017
Classe_Age_Situ_Cont_> 5 ans	-0.0402	0.058	-0.690	0.490	-0.154	0.074
Mode_gestion_P	0.2653	0.045	5.867	0.000	0.177	0.354
Fractionnement_S	0.0639	0.043	1.486	0.137	-0.020	0.148
Fractionnement_T	0.0017	0.039	0.043	0.966	-0.075	0.079
franchise_2. 1 - 200	-0.0543	0.108	-0.501	0.616	-0.267	0.158
franchise_3. 201 - 300	-0.0834	0.110	-0.761	0.447	-0.298	0.131
franchise_4. 301 - 400	-0.1520	0.114	-1.336	0.182	-0.375	0.071
franchise_5. 401 - 600	-0.1454	0.133	-1.092	0.275	-0.406	0.116
franchise_6. > 600	-0.1446	0.146	-0.988	0.323	-0.431	0.142

```
GLM Train Deviance: 7053.970299524621
GLM Validation Deviance: 1451.4090736352814
GLM Train Deviance (normalized): 0.10108581438658423
GLM Validation Deviance (normalized): 0.08319437542332234
```

The Poisson GLM shows very low pseudo R^2, indicating limited explanatory power, which is typical for sparse count data. However, the normalized deviance is relatively low and consistent between training and validation sets, suggesting that the model captures part of the underlying structure without overfitting.

The ratio of Pearson chi-square to residual degrees of freedom is close to one, indicating no strong evidence of overdispersion and supporting the suitability of the Poisson assumption. Nevertheless, several coefficients are not statistically significant and some numerical instability is observed.

Overall, while the model appears limited in terms of explanatory power, it still serves as a useful benchmark for comparison with more flexible models.

1.2 Neural Network

Unlike GLMs, neural networks are sensitive to the scale of input features because they rely on gradient-based optimization. Therefore, all input variables are standardized using a StandardScaler to ensure efficient and stable training.

Four neural network architectures are tested to explore the trade-off between model complexity and generalization (using Keras). The goal is to assess whether increasing depth and capacity improves performance, and whether regularization helps prevent overfitting.

Model 1 serves as the baseline and consists of a small neural network with a single hidden layer of 16 neurons. It is designed to evaluate whether introducing a simple nonlinear structure already provides an improvement over a generalized linear model.

Model 2 increases the level of complexity by using two hidden layers with 32 and 16 neurons, respectively, allowing the network to capture more intricate nonlinear relationships and interactions between features.

Model 3 builds on the same architecture as Model 2 but incorporates dropout, with the aim of assessing whether overfitting is present and whether regularization can improve generalization.

Finally, **Model 4** represents a higher-capacity network with two larger layers of 64 and 32 neurons, intended to test whether greater expressive power leads to better predictive performance or instead results in overfitting.

Model evaluation

We evaluate model performance using the Poisson deviance, which is appropriate for count data.

Table 2: Comparison of training and validation deviances for the neural network models

	Model	Train Deviance	Validation Deviance
0	Model 1	6146.713102	1235.569254
1	Model 2	7201.964206	1599.274014
2	Model 3	5895.765878	1226.990270
3	Model 4	7507.945408	1724.812871

Model 3 achieves the lowest validation deviance (1226.99), making it the best-performing model. Its improvement over Model 2 indicates that incorporating dropout effectively reduces overfitting and enhances generalization.

Model 1 also performs well (1235.57), with validation performance close to that of Model 3. This suggests that even a relatively simple neural network can capture meaningful nonlinear patterns in the data. However, the slight advantage of Model 3 indicates that a moderate increase in complexity combined with regularization can yield additional gains.

In contrast, Model 2 performs worse (1599.27), suggesting that increasing model complexity without regularization leads to overfitting and poorer generalization. Model 4, despite having the highest capacity, shows the worst validation performance (1724.81), reinforcing that simply adding more parameters does not improve results and can significantly degrade generalization.

Compared to the GLM (validation deviance 1451.41), both Model 1 and Model 3 provide clear improvements, demonstrating the benefit of capturing nonlinear relationships. However, the strong performance of the GLM relative to the more complex models (Models 2 and 4) highlights its robustness as a baseline.

Overall, the results show that a moderately complex neural network with appropriate regularization (**Model 3**) provides the best trade-off between flexibility and generalization, while excessive complexity without proper constraints leads to overfitting.

1.3 Subgroup analysis

To assess model calibration beyond global metrics, we compare observed and predicted claim frequencies for both the Generalized Linear Model (GLM) and the selected neural network model (Model 3). Claim frequency is defined as the ratio of total claims to total exposure:

$$\frac{\sum y_i}{\sum e_i} \quad \text{and} \quad \frac{\sum \hat{\mu}_i}{\sum e_i}$$

where y_i denotes observed claims, $\hat{\mu}_i$ predicted claims, and e_i the exposure.

In addition, a multiplicative bias adjustment is applied to assess whether discrepancies arise from a global scaling issue or from structural misspecification. The adjusted predictions are defined as:

$$\hat{\mu}_{\text{adj}} = c\hat{\mu}, \quad c = \frac{\sum y_i}{\sum \hat{\mu}_i}$$

This adjustment ensures that total predicted claims match the observed total, thereby achieving perfect global calibration by construction.

To enable consistent comparison across segments, the validation dataset is reconstructed by combining the original explanatory variables with observed claims, exposure, and model predictions (GLM and neural network). This guarantees proper alignment at the observation level, which is essential for reliable subgroup analysis.

Global and group comparison

Table 3: Global comparison before and after bias adjustment

	Observed	GLM (raw)	NN (raw)	GLM (adjusted)	NN (adjusted)
0	0.111936	0.119955	0.129228	0.111936	0.111936

Both models slightly overestimate the global claim frequency, with a larger deviation observed for the neural network. After applying the bias adjustment, global calibration is achieved by construction, meaning that the predicted and observed totals coincide. This indicates that, on average across the portfolio, the overall level of risk is correctly captured.

To facilitate comparison, exposure is discretized into categories.

Table 4: Comparison by exposure (raw and adjusted predictions)

	exp_group	Observed freq	glm_pred	nn_pred	glm_adj	nn_adj	Count
0	0-0.25	0.118822	0.137930	0.180060	0.128709	0.155967	7594.0
1	0.25-0.5	0.114827	0.128589	0.130231	0.119993	0.112805	4049.0
2	0.5-0.75	0.115638	0.119472	0.124741	0.111485	0.108050	2219.0
3	0.75-1	0.106600	0.109789	0.112389	0.102450	0.097351	3584.0

Table 5: Comparison by geographic area (Zone) (raw and adjusted predictions)

	Zone	Observed freq	glm_pred	nn_pred	glm_adj	nn_adj	Count
0	1	0.088122	0.100915	0.104476	0.094169	0.090497	4158.0
1	2	0.111408	0.111389	0.121834	0.103943	0.105531	3365.0
2	3	0.112645	0.123176	0.136322	0.114942	0.118081	6405.0
3	4	0.122884	0.134246	0.146570	0.125272	0.126958	1579.0
4	5	0.148489	0.146906	0.151800	0.137086	0.131488	1738.0
5	6	0.128911	0.161229	0.157303	0.150451	0.136255	201.0

Table 6: Comparison by activity (raw and adjusted predictions)

	Activite	Observed freq	glm_pred	nn_pred	glm_adj	nn_adj	Count
0	1	0.131077	0.122321	0.131083	0.114144	0.113543	4203.0
1	2	0.094399	0.107132	0.118761	0.099971	0.102870	539.0
2	3	0.089260	0.116113	0.125612	0.108351	0.108805	3238.0
3	4	0.094991	0.123032	0.133433	0.114808	0.115579	1552.0
4	5	0.097770	0.115520	0.123974	0.107798	0.107386	544.0
5	6	0.084180	0.117746	0.124495	0.109874	0.107836	2192.0
6	7	0.128419	0.123121	0.133905	0.114890	0.115988	3318.0
7	8	0.130200	0.120992	0.129821	0.112903	0.112450	1860.0

Across all segments, both models reproduce the general structure of risk but tend to overestimate claim frequency, particularly for low-exposure and higher-risk groups. This effect is more pronounced for the neural network.

While the bias adjustment corrects the global level, discrepancies persist across subgroups. This indicates that the miscalibration is not solely due to a scaling issue, but also reflects structural differences in how the models capture risk. The neural network exhibits greater variability across segments, whereas the GLM provides more stable and consistent estimates, likely due to its explicit incorporation of exposure through the offset.

Overall, in this setting, the additional flexibility of the neural network does not appear to result in improved calibration compared to the GLM.

1.4 Determining the Most Discriminative Factors (PDP & ICE)

To identify the variables that most strongly influence claim frequency, we rely on Partial Dependence Plots (PDP) and Individual Conditional Expectation (ICE). These complementary tools allow us to analyze how model predictions evolve as a function of a given feature, either on average (PDP) or at the individual level (ICE), thereby capturing both global and heterogeneous effects.

The analysis is conducted on a representative subsample (2000 observations) of the data to ensure computational efficiency while preserving interpretability. The most influential features are first identified using **permutation-based importance**, after which the top variables are examined in detail using PDP and ICE plots to assess their marginal effects. The ranking of the most important features is summarized below.

Table 7: Most influential predictors for the neural network model

	Feature	Importance
9	duration_days	1.204308
6	ValeurPuissance	0.194269
17	Mode_gestion_P	0.173565
21	franchise_3. 201 - 300	0.141038
20	franchise_2. 1 - 200	0.116533

PDP and ICE Analysis

Figure 1 presents the PDP and ICE plots for the most influential variables identified previously. These plots illustrate both the average effect of each feature on predicted claim frequency and the variability of this effect across individuals. To complement the visual analysis, Table 8 reports the range of variation of the PDP for each feature.

Figure 1: PDP and ICE plots for selected features

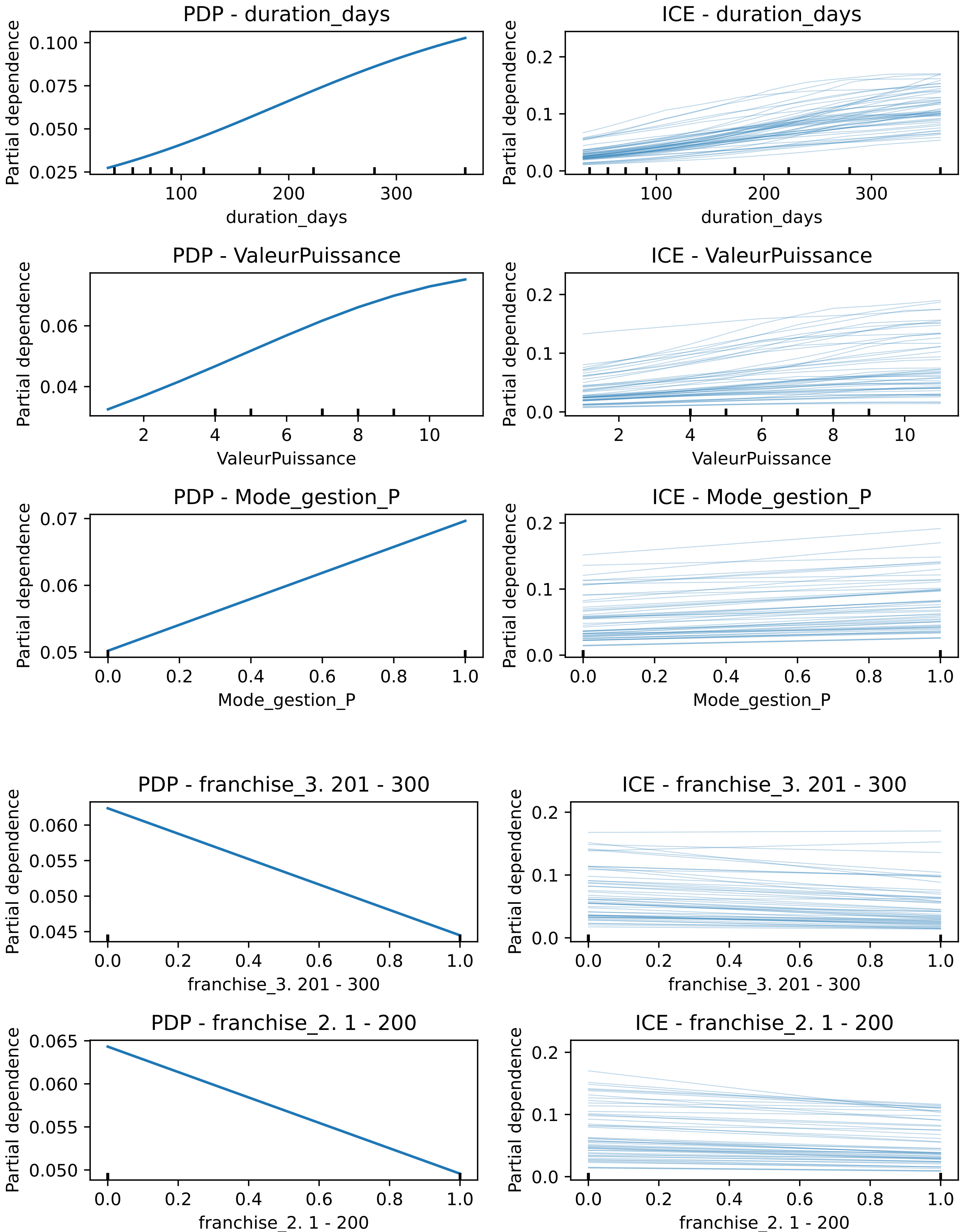


Table 8: Influence of each feature by PDP Range

	Feature	PDP Range
0	duration_days	0.075317
1	ValeurPuissance	0.042739
2	Mode_gestion_P	0.019449
3	franchise_3. 201 - 300	0.017887
4	franchise_2. 1 - 200	0.014757

The results reveal a clear hierarchy among the most influential variables in the model. Contract duration (`duration_days`) emerges as the dominant feature, with the largest PDP range. This indicates that the predicted claim frequency increases substantially with the length of the contract, highlighting the central role of exposure in driving risk.

Vehicle power (`ValeurPuissance`) is the second most influential variable, showing a strong positive relationship with claim frequency. This suggests that higher-powered vehicles are associated with systematically higher risk, making this variable particularly relevant for risk segmentation and pricing.

Overall, the results indicate that claim frequency is primarily driven by exposure-related factors and key vehicle characteristics. Variables such as contract duration and vehicle power provide strong and stable signals for risk differentiation, making them particularly suitable for pricing decisions. In contrast, variables with smaller effects contribute to refining the model but play a more limited role in distinguishing high- and low-risk policyholders.

1.5 LIME and SHAP

To complement the global interpretability analysis (PDP and ICE), we perform a local analysis using LIME and SHAP. These approaches provide instance-level explanations, allowing us to identify the variables driving predictions for individual policies. Two representative observations are selected based on the predictions of the best-performing neural network model (Model 3): one with the lowest predicted claim frequency (low-risk policy) and one with the highest predicted claim frequency (high-risk policy).

LIME analysis

LIME provides local explanations by expressing each prediction as a set of feature contributions. Positive contributions increase the predicted claim frequency, while negative contributions decrease it.

Table 9: LIME explanation for low-risk policy

	Condition	Contribution	Impact
0	year_start <= 2011.00	-0.034592	↓ Decrease
1	duration_days <= 61.00	-0.015211	↓ Decrease
2	Mode_gestion_P <= 0.00	-0.009723	↓ Decrease

	Condition	Contribution	Impact
3	franchise_6. > 600 <= 0.00	0.008475	↑ Increase
4	0.00 < franchise_3. 201 - 300 <= 1.00	-0.008200	↓ Decrease
5	franchise_2. 1 - 200 <= 0.00	0.006696	↑ Increase
6	Creation_Entr <= 1.00	-0.006513	↓ Decrease
7	ValeurPuissance <= 4.00	-0.006359	↓ Decrease
8	franchise_5. 401 - 600 <= 0.00	0.005005	↑ Increase
9	1.00 < Segment <= 2.00	0.004867	↑ Increase

Table 10: LIME explanation for high-risk policy

	Condition	Contribution	Impact
0	duration_days > 258.00	0.024324	↑ Increase
1	2012.00 < year_start <= 2013.00	0.017042	↑ Increase
2	0.00 < Mode_gestion_P <= 1.00	0.009857	↑ Increase
3	franchise_6. > 600 <= 0.00	0.009606	↑ Increase
4	franchise_2. 1 - 200 <= 0.00	0.007509	↑ Increase
5	franchise_3. 201 - 300 <= 0.00	0.007182	↑ Increase
6	franchise_4. 301 - 400 <= 0.00	0.005744	↑ Increase
7	7.00 < ValeurPuissance <= 8.00	0.005248	↑ Increase
8	Segment <= 1.00	-0.005031	↓ Decrease
9	1.00 < Type_Apporteur <= 2.00	0.005012	↑ Increase

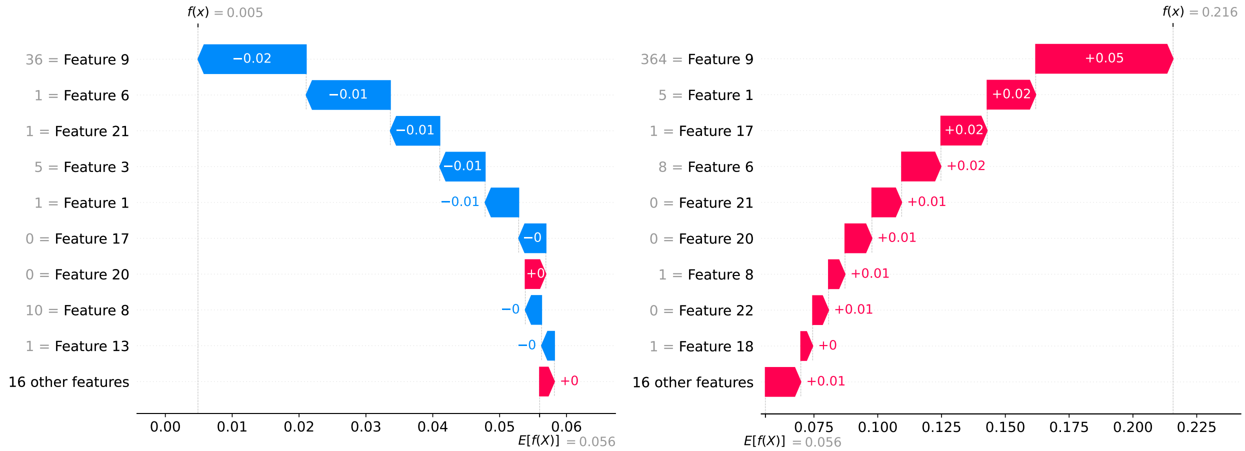
For the **low-risk policy**, the prediction is primarily driven downward by several negative contributions. Short contract duration (`duration_days<=61`) is one of the strongest contributors, significantly reducing the predicted claim frequency. Earlier policy start years (`year_start<=2011`) and lower vehicle power (`ValeurPuissance<=4`) also contribute to lowering the prediction. Other factors, such as management mode and certain franchise levels, further reinforce this downward effect. Although some franchise-related variables contribute positively to the prediction, their impact remains limited and is outweighed by the dominant negative contributions, resulting in a low overall predicted risk.

For the **high-risk policy**, the prediction is largely driven upward by positive contributions. A long contract duration (`durations_days>258`) is the strongest contributor, reflecting increased exposure. Policy start years in the range 2012–2013 also contribute positively to the prediction. This effect is reinforced by additional factors such as higher vehicle power, management structure, and several franchise-related variables. A small negative contribution is observed for the fleet size variable (`Segment<=1`) but this is not sufficient to offset the cumulative positive effects, leading to a high predicted claim frequency.

SHAP analysis

SHAP provides an additive decomposition of each prediction relative to a baseline value, allowing precise quantification of feature contributions.

(a) 1 - Low-risk policy (a) 2 - High-risk policy



SHAP Waterfall Plots

Table 11: Feature Contribution Comparison

	Feature	Policy_1_SHAP	Policy_2_SHAP	Difference
9	duration_days	-0.016185	0.053819	-0.070004
6	ValeurPuissance	-0.012583	0.015393	-0.027976
1	Zone	-0.005021	0.019026	-0.024047
17	Mode_gestion_P	-0.004072	0.018125	-0.022197
21	franchise_3. 201 - 300	-0.007396	0.011647	-0.019043
3	FORMULE	-0.006782	0.002532	-0.009314
8	month_start	-0.002434	0.006442	-0.008876
20	franchise_2. 1 - 200	0.003085	0.010549	-0.007464
18	Fractionnement_S	-0.001037	0.004650	-0.005687
22	franchise_4. 301 - 400	0.000639	0.006217	-0.005578

For the Shapley waterfall plots, the results can be interpreted as follows. For the **low-risk policy**, the predicted claim frequency (0.005) is well below the baseline (0.056), primarily due to negative feature contributions. This indicates that, according to the model, the characteristics of this policy collectively push the prediction toward a lower-than-average risk.

In contrast, for the **high-risk policy**, the predicted claim frequency (0.216) is substantially above the same baseline (0.056), driven by strong positive feature contributions. This suggests that the model associates the policy's characteristics with a higher-than-average level of risk.

The comparison of SHAP values between the two policies provides further insight into the drivers of this gap. The difference is primarily explained by contract duration (*duration_days*, -0.075). Other important differentiating features include ValeurPuissance and Zone (*both around* -0.026). These feature decrease the predicted risk for Policy 1 while substantially increasing it for Policy 2, making it the main contributor to the prediction difference.

Additionally, franchise_5 contributes more positively to the prediction in Policy 2, further reinforcing its higher predicted risk, although its effect remains smaller compared to the main drivers.

Comparison between LIME and SHAP

Both LIME and SHAP identify contract duration, ValeurPuissance, and contract characteristics as the main drivers of claim frequency, confirming the consistency of the model's behavior. Their interpretations differ in emphasis:

SHAP provides a comparative and additive view, explaining how each feature contributes to differences in predicted frequency between policies and highlighting contract duration as the key factor separating low-risk and high-risk profiles, with opposite effects.

LIME, by contrast, offers a local and more narrative explanation of individual predictions, showing how feature values cumulatively push each policy toward lower or higher risk, with low-risk policies driven by factors reducing frequency and high-risk policies by factors increasing it.

Overall, both approaches indicate that risk differentiation is primarily driven by contract structure, particularly duration. # 2. Unsupervised learning

2.1 Autoencoders & Variational Autoencoders

The dataset is first prepared for unsupervised learning by selecting a subset of relevant features describing contracts (`FORMULE`, `Fractionnement` and `Mode_gestion`), vehicle characteristics (`ValeurPuissance`), risk segmentation variables (`Segment`,`Zone` and `Activite`) and portfolio structure (`Type_Apporteur` and `Creation_Entr`).

All categorical variables are then transformed using one-hot encoding, allowing neural networks to process them appropriately and missing values are replaced by zeros.

Finally, the data is standardized using a *StandardScaler*, ensuring that all features are on comparable scales and improving convergence during training. The resulting dataset has a:

```
Shape of (87228, 10)
```

Model training: AE's and VAE's

Four models are trained: two standard autoencoders (AE1 and AE2) and two variational autoencoders (VAE1 and VAE2), designed to compare reconstruction performance, model capacity, and latent space structure.

AE1 is a simple, symmetric autoencoder with a single hidden layer before and after a **low-dimensional bottleneck (8 units)**. The encoder compresses the input into this compact representation, and the decoder reconstructs the original input from it. This model serves as a **baseline**, testing how well the data can be reconstructed under limited model capacity.

AE2 increases both the width and depth of the network, using a larger hidden representation and a **moderate latent space (16 units)**. It also incorporates **dropout** in the encoder, introducing stochastic regularization during training. This model tests whether increased capacity improves reconstruction quality.

VAE1 replaces the deterministic latent space with a **probabilistic representation of very low dimensionality (2 dimensions)**. Instead of encoding each input as a single point, the model learns a distribution (mean and variance) and samples from it. The training objective combines reconstruction loss with a **KL divergence term**, which regularizes the latent space to follow a standard normal distribution.

VAE2 extends VAE1 by increasing both **network depth and latent dimensionality (4 dimensions)**. This provides greater flexibility in encoding complex patterns while still enforcing a structured latent distribution. This model tests whether additional latent capacity improves reconstruction while maintaining a well-regularized latent space.

Model Evaluation

Model performance is evaluated using the **mean squared reconstruction error (MSE)**, which measures how accurately each model reproduces the original input data.

The results are summarized below:

Table 12: Reconstruction performance (MSE) of AE and VAE models

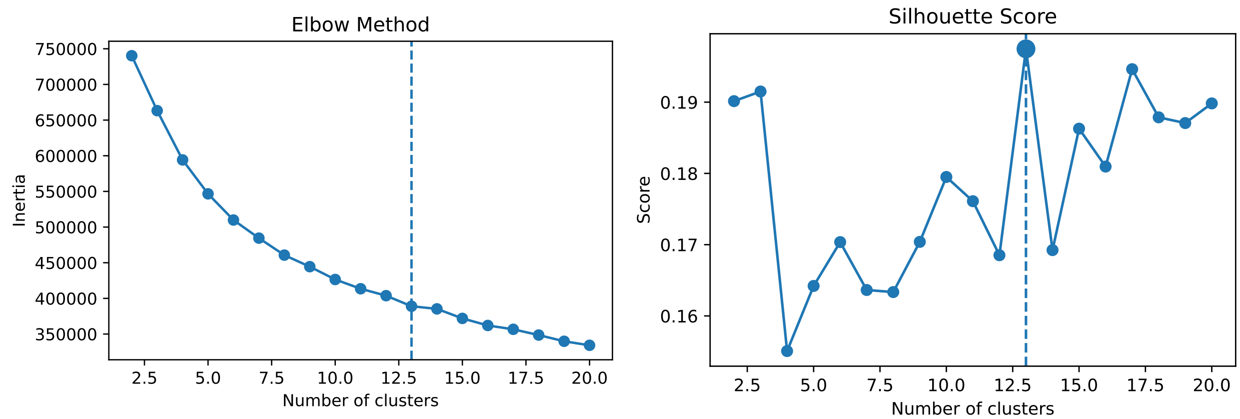
	Model	Reconstruction MSE
0	AE1	0.001469
1	AE2	0.018809
2	VAE1	1.002411
3	VAE2	1.001987

The results show that AE1 achieves the lowest reconstruction error, indicating strong performance in compressing and reconstructing the data under a tight latent constraint. AE2 exhibits higher reconstruction error, which can be attributed to both its increased complexity and the use of dropout regularization.

In contrast, the variational autoencoders (VAE1 and VAE2) show substantially higher reconstruction errors. This behavior is expected, as VAEs optimize a combined objective that includes both reconstruction loss and a Kullback–Leibler (KL) divergence term.

However, the magnitude of the difference suggests that the balance between reconstruction and regularization may be skewed toward the KL term. Between the two VAE models, VAE2 performs slightly better, indicating that increasing the latent dimensionality provides additional flexibility for encoding the data.

Overall, these results suggest that while standard autoencoders provide better reconstruction performance, the additional regularization introduced by variational autoencoders may be beneficial for capturing more robust latent representations, which could be advantageous in the context of sparse and heterogeneous insurance data.



2.2 K means algorithm

We apply the K-means algorithm to identify clusters of insured individuals. The number of clusters is optimized (with a maximum of 20), and the characteristics of each cluster are subsequently analyzed.

Based on the silhouette score, the optimal number of clusters is 13. The elbow method provides a consistent but less distinct indication.

Clustering results

Table 13: Cluster sizes

	C0	C1	C2	C3	C4	C5	C6	C7	C8	C9	C10	C11	C12
count	3644	14101	5284	1640	4842	10106	5943	6085	8448	9998	7137	4478	5522

Cluster sizes vary substantially across groups, with Cluster 1 being the largest (14101 observations) and Cluster 3 the smallest (1640 observations). This heterogeneity suggests that the portfolio is not evenly segmented, with some profiles being much more common than others.

Furthermore, the most important features within each cluster are identified based on their average values. For one-hot encoded variables, these averages correspond to category proportions and directly indicate dominant characteristics. For numerical variables, they reflect relative differences between clusters.

Table 14: Summary table of clusters

Cluster	0	1	2	3	4	5	6	7	8	9	10	11	12
Type_Apporteur	1.14	1.15	1.11	2.98	1.47	1.15	1.95	1.36	1.13	1.23	1.22	2.82	1.13
Creation_Entr	2.00	1.00	1.00	2.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	2.00
ValeurPuissance	6.16	6.41	5.97	6.04	6.34	6.42	6.13	6.21	6.09	6.32	6.07	5.89	6.35
Activite	1.29	4.84	4.50	3.66	4.71	4.88	4.59	4.53	4.50	4.91	4.32	3.87	2.41
FORMULE	5.00	3.00	3.00	4.00	3.00	3.00	3.44	3.00	3.00	3.00	5.00	4.18	3.00

Cluster	0	1	2	3	4	5	6	7	8	9	10	11	12
Zone	2.72	2.58	2.43	2.85	2.81	2.53	3.03	2.66	2.39	2.81	2.33	3.06	2.69
Segment	1.52	1.00	2.00	1.82	1.72	1.00	1.84	1.72	2.00	1.34	1.56	1.72	1.46
Fractionnement_S	0.35	0.00	0.00	0.38	0.00	0.00	1.00	0.00	0.00	1.00	0.24	0.00	0.28
Fractionnement_T	0.27	1.00	0.00	0.25	0.00	0.00	0.00	1.00	1.00	0.00	0.35	0.63	0.38
Mode_gestion_P	0.86	0.00	0.00	0.96	1.00	0.00	0.99	1.00	0.00	0.04	0.08	0.96	0.60

Table 15: Dominant features per cluster

	Cluster	Top 1	Top 2	Top 3	Top 4	Top 5
0	0	ValeurPuissance	FORMULE	Zone	Creation_Entr	Segment
1	1	ValeurPuissance	Activite	FORMULE	Zone	Type_Apporteur
2	2	ValeurPuissance	Activite	FORMULE	Zone	Segment
3	3	ValeurPuissance	FORMULE	Activite	Type_Apporteur	Zone
4	4	ValeurPuissance	Activite	FORMULE	Zone	Segment
5	5	ValeurPuissance	Activite	FORMULE	Zone	Type_Apporteur
6	6	ValeurPuissance	Activite	FORMULE	Zone	Type_Apporteur
7	7	ValeurPuissance	Activite	FORMULE	Zone	Segment
8	8	ValeurPuissance	Activite	FORMULE	Zone	Segment
9	9	ValeurPuissance	Activite	FORMULE	Zone	Segment
10	10	ValeurPuissance	FORMULE	Activite	Zone	Segment
11	11	ValeurPuissance	FORMULE	Activite	Zone	Type_Apporteur
12	12	ValeurPuissance	FORMULE	Zone	Activite	Creation_Entr

Vehicle power (`ValeurPuissance`) is the most discriminating variable across all clusters, suggesting it plays a key role in structuring the segmentation. Other important variables include contract characteristics (`FORMULE` and `Activite`), geographical location (`Zone`), and policyholder type (`Type_Apporteur`).

Several distinct profiles emerge. Some clusters are associated with higher activity levels and standard contract formulas, indicating typical usage patterns. Others differ in terms of policyholder type or geographical concentration, suggesting variations in distribution channels or risk environments.

Overall, the analysis shows that the motor fleet portfolio is highly heterogeneous, with 13 distinct clusters capturing diverse risk profiles. Vehicle power emerges as the key differentiating factor, alongside contract features, location, and policyholder type. This segmentation highlights that claim frequency behavior is not uniform across the portfolio.

2.3 GLM vs Neural Networks vs Clusters

This section compares four approaches for modeling claim frequency: a Generalized Linear Model (GLM), a Neural Network (NN), a cluster-based model derived from K-means, and a baseline model based on the global claim frequency.

The dataset is preprocessed by removing non-informative variables, encoding categorical features using one-hot encoding, and imputing missing values. Exposure is consistently incorporated across models: as an offset in the GLM, as sample weights in the Neural Network, and as a scaling factor in the cluster-based approach.

The cluster-based model introduces a segmentation of the portfolio by grouping similar observations using K-means and estimating claim frequencies at the cluster level. A smoothing procedure is applied to improve the stability of estimates, particularly for smaller clusters. In contrast, the baseline model assumes a homogeneous portfolio and assigns the same global frequency to all observations.

Model performance is evaluated on a **validation set** using the Poisson deviance, allowing for a consistent comparison of predictive accuracy across approaches.

Table 16: Model performance evaluation

	Model	Deviance	Normalized Deviance
1	Neural Network	1235.569254	0.070822
0	GLM	1451.409074	0.083194
2	Clusters	1525.055964	0.087416
3	Baseline	1543.869272	0.088494

The Neural Network achieves the lowest deviance, indicating the best predictive performance, although the improvement over the GLM remains modest (around 2%). This suggests that nonlinear effects are present but limited in magnitude.

The GLM performs nearly as well, confirming that a log-linear specification captures most of the relationship between predictors and claim frequency. Given its interpretability and robustness, it remains a strong benchmark model.

The cluster-based model performs worse than both supervised approaches, although it still improves upon the baseline. The baseline model yields the highest deviance, highlighting the importance of incorporating explanatory variables and accounting for heterogeneity in the portfolio.

Overall, the results show that accurately modeling claim frequency significantly benefits from incorporating policyholder characteristics, confirming the importance of accounting for portfolio heterogeneity in motor fleet insurance. While the Neural Network provides the best predictive performance, its limited improvement over the GLM suggests that most of the underlying risk structure can be effectively captured through a well-specified log-linear model. Consequently, the GLM appears to offer the best trade-off between accuracy, interpretability, and robustness, making it particularly well-suited for practical pricing applications.

Conclusion

This study aimed to model claim frequency in a motor insurance portfolio while accounting for exposure, and to compare the effectiveness of several analytical approaches. The results demonstrate that a Poisson Generalized Linear Model (GLM) provides a strong and reliable benchmark, combining robustness, stability, and interpretability despite the sparsity and imbalance inherent in the data.

Neural networks introduce additional flexibility and are able to achieve slight improvements in predictive performance when properly regularized. However, these gains remain limited, suggesting that the relationship between predictors and claim frequency is largely well captured by a log-linear structure. Moreover, the increased complexity of neural networks comes at the cost of reduced interpretability and less stable calibration across portfolio segments.

The unsupervised learning analysis reveals significant heterogeneity within the portfolio. Autoencoders successfully learn compact representations of the data, while clustering methods identify distinct groups of policyholders characterized by differences in contract features, vehicle attributes, and geographic exposure. Nevertheless, these approaches are less effective for direct prediction, confirming that their primary value lies in segmentation, exploration, and portfolio understanding rather than predictive accuracy.

Across all methods, several key drivers of claim frequency consistently emerge, particularly contract duration and vehicle power. These variables provide actionable insights for risk assessment and pricing.

Overall, the results highlight that increased model complexity does not necessarily translate into substantial performance gains in this context. Instead, simpler and well-specified models such as the GLM remain highly competitive and operationally advantageous. Future work could focus on improving calibration across subgroups, exploring alternative count models (e.g., Negative Binomial models), or incorporating richer features to better capture residual heterogeneity.

In conclusion, combining interpretable statistical models with complementary machine learning and segmentation techniques provides a more comprehensive understanding of insurance risk and supports more informed and balanced decision-making.